higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(Second Paper)

NQF LEVEL 4

(12050674)

15 March 2019 (Y-Paper)
13:00–15:00

REQUIREMENTS: **A4 drawing sheets**
Yellow cover folder (BOE 8/15)
Standard hot-rolled structural steel section table (BOE 8/1)

Calculators and drawing instruments may be used.

This question paper consists of 4 pages.

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TIME: 2 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Use a suitable scale if NOT indicated.
6. Make drawings and sketches clear enough to show detail.
7. Fully dimension and label ALL drawings.
8. Insert ALL loose DRAWING SHEETS in the folder and do NOT staple.
9. Work neatly.

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QUESTION 1: CONCRETE RETAINING WALL

1.1 Draw a concrete retaining wall using CAD. The wall extends 280 mm above the ground. The larger mass of soil has natural a slope of 20° and the bottom soil is level. The toe is situated on the right-hand side of the wall.

Use a scale of 1:20 with the following information:

Height of the wall:	2 500 mm
Wall thickness:	0,5 m
Length of the toe:	1 500 mm
Length of Tail	600mm
Thickness of the base:	0,6 m
Concrete cover:	60 mm
Wall longitudinal reinforcement:	Y12 @ 150 c/c and transversely with Y16s @ 300 c/c
Base longitudinal reinforcement:	Y16 @ 150 c/c and transversely with Y20s @ 400 c/c

(20)

1.2 Draw, to scale 1:10, a cross section through a double-reinforced beam. The beam consists of concrete, stirrups, SIX compression bars and SIX tension bars. The drawing must include the stirrups, the compression rods, the tension rods and the concrete cover.

Reinforced beam:	800 mm × 870 mm
Tensile rods:	20 mm
Compression rods:	20 mm
Stirrups:	10 mm diameter
Concrete cover:	50 mm

(15)
[35]

QUESTION 2: STEEL STAIRCASE

Draw, to scale 1:5, a steel staircase using CAD. The rise is 100 mm and the going is 200 mm.

Label ALL of the components indicated in the following table:

Number	Component	Size
1	Vertical head	Any suitable height
2	Going	200 mm
3	Rise	110 mm
4	Pitch angle	31°
5	Overlap	34 mm
6	Landing	400 mm

[20]

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QUESTION 3: STRUCTURAL STEEL WORK DETAILING

3.1 A 203 × 203 × 53,5 kg/m steel H-profile with parallel flanges is used as a column and two 457 × 191 × 67 kg/m steel I-profiles with parallel flanges are used as the main beam connected to the web of the column. One 254 × 146 × 31,3 kg/m steel I-profile with parallel flanges is connected as a secondary beam to the flange of the column. ALL beams are on line at the top surface of the beam.

3.1.1 Draw the secondary beam facing the front view in first-angle orthographic projection.

3.1.2 Draw the main beam facing the right view.

3.1.3 Draw the main beam facing the top view.

Select any suitable scale, draw ALL views in the correct position, name ALL views and label ALL views. (20)

[20]

TOTAL: 75

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